

Back Week 2 Quiz Graded Quiz

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- Creating Convolutional Neural Networks in JavaScript
Using a Sprite Sheet
MNIST Classifier
- Video: Using tf.tidy() to Save Memory

1 min
- Video: A Few Words From Laurence

58 sec
- Reading: MNIST Classifier

10 min
- Video: MNIST Classifier In Code

13 min
- Quiz: Week 2 Quiz

Submitted
- Reading: Week 2 Wrap up

10 min
- Lecture Notes (Optional)
Graded Exercise: Fashion MNIST Classifier

Week 2 Quiz

Submit your assignment

Due Dec 31, 11:59 PM IST Attempts 3 every 8 hours

Receive grade

To Pass 80% or higher

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Congratulations! You passed!

Grade received 87.50% Latest Submission Grade 87.50% To pass 80% or higher Go to next item

Try again

1. What is the correct syntax for the first layer in a convolutional neural network that takes an MNIST (28x28 monochrome) input?

1 / 1 point

1 model.add(tf.layers.conv2d([inputShape: [28, 28, 1], kernelSize: 3, filters: 8, activation: 'relu']));

Your grade 87.50% View Feedback We keep your highest score

1 model.add(tf.layers.conv([inputShape: [28, 28, 1], kernelSize: 3, filters: 8, activation: 'relu']));

1 model.add(tf.layers.conv2d([inputShape: [28, 28], kernelSize: 3, filters: 8, activation: 'relu']));
2

1 model.add(tf.layers.conv([inputShape: [28, 28, 1], kernelSize: 3, filters: 8, activation: 'relu']));

Correct

2. What is the correct syntax for adding a maxPooling2D layer to a Convolutional neural network in JavaScript?

1 / 1 point

1 model.add(tf.layers.maxPooling2D(poolSize: [2, 2]));

1 model.add(tf.layers.maxPooling2d(poolSize: [2, 2]));

1 model.add(tf.layers.maxPooling2d(poolSize = [2, 2]));

1 model.add(tf.layers.maxPooling2D(poolSize = [2, 2]));

Correct

3. What is the correct syntax for compiling a model with an optimizer, loss function and metrics?

1 / 1 point

model.compile({ optimizer: tf.train.adam(), loss: 'categoricalCrossentropy', metrics: ['accuracy']});

1 model.compile({ optimizer = tf.train.adam(), loss = 'categoricalCrossentropy', metrics = ['accuracy']});

1 model.compile({ tf.optimizer: tf.train.adam(), tf.loss: 'categoricalCrossentropy', tf.metrics: ['accuracy']});

1 model.compile({ optimizer: tf.train.adam(); loss: 'categoricalCrossentropy'; metrics: ['accuracy']});

Correct

4. How do you correctly pass a set of validation data called testXs and testYs to the model.fit method in JavaScript?

1 / 1 point

- Use validationData = [testXs, testYs] in the list of parameters to model.fit
- Use validationData: [testXs, testYs] in the list of parameters sent as the third parameter to model.fit
- Use validationData: [testXs, testYs] in the list of parameters to model.fit
- Use validationData= [testXs, testYs] and pass it to the model.fit method

Correct

5. How do you get the built in callbacks visualizer with TensorFlow.js?

0 / 1 point

- Include the tfjs-vis script, set a callback in model.fit and it will work automatically
- Include the tfjs-vis script and it will work automatically
- Include the tfjs-vis script, call show.fitCallbacks() on the tfvis object
- Include the tfjs-vis script, set a callback in model.fit, and set it to a const that called show.fitCallbacks() on the tfvis object

Incorrect

6. If you want to see loss, validation loss, accuracy and validation accuracy on each epoch while training, how do you do this?

1 / 1 point

- Create a list containing text values with the names of the analytics you want to capture, i.e. ['loss', 'val_loss', 'acc', 'val_acc'] and pass it to fitCallbacks() as a parameter
- Create a list containing [1, 1, 1, 1] indicating that you want those 4 values to be true and pass it to the fitCallbacks() as a parameter
- Create a list setting loss=true, val_loss=true, acc=true, val_acc=true, and pass it to the fitCallbacks() as a parameter
- Create a list containing text values ["loss=true", "val_loss=true", "acc=true", "val_acc=true"] and pass it to fitCallbacks() as a parameter

Correct

7. When using a dataset like MNIST or FashionMNIST, why is it advisable to use a sprite sheet containing all the images?

1 / 1 point

- It prevents excessive multiple HTTP calls to download the data
- It keeps the data in the native JS format
- It doesn't require any additional pre-processing
- It makes the data more secure

Correct

8. What is the role of tf.tidy() in TensorFlow.js?

1 / 1 point

- When it is executed, it removes everything tensorflow from the browser memory and cache
- When it is executed, it cleans up all intermediate tensors allocated by a function except those returned by the function
- It shuts down tensorflow when done, cleaning up all memory
- When it is executed it clears memory for new tensors

Correct